



KARAN PRATAP SINGH RATHORE

✉ karanpratap1810@gmail.com

☎ +91-7413087071

🌐 rathorekaran.site

🐙 github.com/KaraNxKP007

🌐 linkedin.com/in/karanpratap7

EDUCATION

Year	Degree/Certificate & Institute/Board	Institute	CGPA/Marks
2022–2026	B. Tech. in Chemical Engineering Indian Institute of Technology Jodhpur	IIT Jodhpur	8.58 / 10
2020–2021	Senior Secondary Examination RBSE	Vyas Public Senior Sec. School Bikaner	99.40%

WORK EXPERIENCE

- Chiral** **Dec 2025**
- Developed a computer vision pipeline using YOLOv8 and OpenCV to automate the detection, classification, and diameter measurement of microscopic organoids, improving analysis accuracy and reducing manual effort.
- Sharp Career Technologies** **Jun 2024 – Jul 2024**
- Designed and developed a full-stack hotel booking web application using React, JavaScript, HTML5, CSS3, Bootstrap, and REST APIs, participating in requirement analysis, feature development, and relational database integration.
 - Built reusable UI components, performed refactoring, testing, and bug fixing, while optimizing performance, accessibility, and cross-browser compatibility.
- IChE Student Chapter, IIT Jodhpur** **Aug 2023 – May 2024**
- Designed, developed, and deployed the official chapter website using HTML, CSS, and JavaScript, achieving a 50-point improvement in Google Lighthouse performance.
 - Led UI/UX implementation and feature rollouts for chapter events, delivering a responsive, mobile-friendly platform that increased student engagement.

PROJECTS

- TalentBridge – Career Platform** |
 - Built a full-stack system design architecture of career platform featuring job listings, recruiter–student matchmaking, resume parsing, and an AI-powered recommendation engine.
 - Designed a scalable three-tier architecture with React.js, Node.js, PostgreSQL, and REST APIs, implementing real-time notifications and secure backend services.
- CodeBulb – AI Pilot for VS Code** | *Dec 2025 – Jan 2026*
 - Built an AI-powered VS Code extension integrating Google Gemini and DeepSeek for real-time code generation, debugging, and project scaffolding.
 - Developed autonomous Node.js agents for file management and shell execution, and published the extension on the VS Code Marketplace with a custom WebView interface.
- Fisher–Yates Algorithm Visualizer** |
 - Developed an interactive single-page application to visualize the Fisher–Yates shuffle algorithm using asynchronous JavaScript animations and dynamic DOM manipulation.
 - Implemented performance reporting with jsPDF and custom regex-based sanitization to generate downloadable execution reports.

COMPETITIONS / CONFERENCES

- Amazon ML Challenge 2025 — Smart Product Pricing** **2025**
- Placed in the **top 1%** in the Smart Product Pricing Challenge (Amazon ML).
 - Engineered a multimodal pricing model combining text embeddings, CLIP-based image features, and metadata, improving SMAPE from a 61–66% baseline to 45–35%, a 20 percentage-point reduction.

TECHNICAL SKILLS

- Programming:** PHP, JavaScript (ES6+), TypeScript, Python, C++, SQL
- Frontend:** React.js, HTML5, CSS3, Bootstrap, Tailwind CSS
- Backend:** Node.js, Express.js, REST APIs, Laravel
- AI & ML:** Generative AI, LLMs, RAG, Prompt Engineering, OpenCV, YOLOv8, Deep Learning, Numpy, Pandas, Pytorch, Tensorflow, Langchain, LangGraph, Matplotlib, Scikit-Learn, Seaborn
- Databases:** MySQL, PostgreSQL, Vector Databases (Pinecone), Graph Databases (neo4j)
- Developer Tools:** Git, GitHub, Linux, Docker, Postman

RELEVANT COURSEWORK

Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Machine Learning, Probability & Statistics, Scientific Computations, Process Control & AI Applications.

POSITIONS OF RESPONSIBILITY

- Coordinator:** Career Guidance Cell, IIT Jodhpur *May 2024 – Apr 2025*
- Developer Head:** IChE Student Chapter, IIT Jodhpur *Sep 2024 – Dec 2024*

AWARDS AND ACHIEVEMENTS

- Graduated with Department **Rank 2**, B.Tech. Chemical Engineering, IIT Jodhpur 2026.